

Linear Algebra I Recitation Class: Week 5

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Throughout these notes, $\mathbb{Q}$, $\mathbb{R}$, and $\mathbb{C}$ denote the sets of rational, real, and complex numbers, respectively. V, W are vector spaces over a field F . If you are not yet familiar with fields, you may keep the examples $F = \mathbb{Q}$, $\mathbb{R}$, or $\mathbb{C}$ in mind.

A linear transformation $T : V \rightarrow W$ is a function that satisfies

1. $T(v + w) = T(v) + T(w)$ for all $v, w \in V$.
2. $T(av) = aT(v)$ for all $v \in V, a \in F$.

and these two conditions can be combined into one:

Exercise A. Prove that $T : V \rightarrow W$ is a linear transformation if and only if $T(v + aw) = T(v) + aT(w)$ for any $a \in F, v, w \in V$.

To save chalk, we say T is linear when T is a linear transformation. Let $T : V \rightarrow W$ be linear, we can define the image $\text{im } T$ of T and the kernel $\ker T$ of T by

$$\text{im } T := T(V) = \{T(v) \mid v \in V\} \quad \ker T := T^{-1}(0_W) = \{v \in V \mid T(v) = 0_W\}.$$

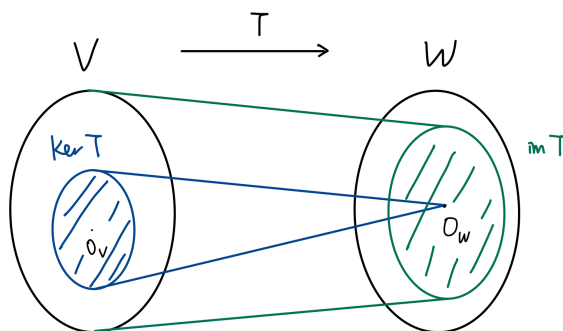


Figure 1: $\text{im } T$ and $\ker T$

Recall that $\text{im } T$ is a subspace of W , and $\ker T$ is a subspace of V . In fact, we can say more about it:

Exercise B. Let $T : V \rightarrow W$ be linear.

1. Prove that for any subspace V' of V , $T(V')$ is a subspace of W .
2. Prove that for any subspace W' of W , $T^{-1}(W')$ is a subspace of V .

Recall that a function $f : V \rightarrow W$ is one to one (or **injective**) if $f(v_1) = f(v_2) \implies v_1 = v_2$ for any $v_1, v_2 \in V$; and f is onto (or **surjective**) if $\text{im } f = W$, that is, for any $w \in W$, $w = f(v)$ for some $v \in V$. When $T : V \rightarrow W$ is linear, we have a useful criterion for injectivity:

Proposition 1. Let $T : V \rightarrow W$ be linear, then T is one-to-one if and only if $\ker T = \{0\}$.

Now we focus on linear transformations from a **finite dimensional** vector space V to a vector space W . Recall that we have a dimension formula

Proposition 2. Let $T : V \rightarrow W$ be linear and V be **finite dimensional**, then

$$\dim V = \dim \ker T + \dim \operatorname{im} T.$$

Exercise C. Explain why $\operatorname{im} T$ is finite dimensional even if we do not assume W to be finite dimensional? (Hint: $T(\operatorname{span}(S)) = \operatorname{span} T(S)$ for any subset $S \subseteq V$.)

Using the dimension formula, we can obtain the dimension for quotient spaces: Let V be a **finite dimensional** vector space and W be a subspace of V , then we define a function

$$\pi : V \longrightarrow V/W \quad v \longmapsto v + W.$$

Given $v, w \in V$, we have

$$\pi(v + w) = (v + w) + W = (v + W) + (w + W) = \pi(v) + \pi(w).$$

For any $v \in V$, $a \in F$, we have

$$\pi(av) = (av + W) = a(w + W) = a\pi(v).$$

Therefore, π is linear. Given $x \in V/W$, $x = v + W$ for some $v \in V$. Hence $\pi(v) = v + W = x$, which proves that π is surjective. Furthermore, since $w + W = W = 0_{V/W}$ if and only if $w \in W$, we have $\ker \pi = W$. Then by the dimension formula

$$\begin{aligned} \dim V &= \dim \ker \pi + \dim \operatorname{im} \pi \\ &= \underbrace{\dim W}_{\ker \pi = W} + \underbrace{\dim V/W}_{\operatorname{im} \pi = V/W} \end{aligned}$$

This proves the following theorem:

Theorem 3. Let V be a **finite dimensional** vector space and W be a subspace of V , the map

$$\pi : V \longrightarrow V/W \quad v \longmapsto v + W.$$

is a surjective linear transformation with $\ker \pi = W$. In particular,

$$\dim V/W = \dim V - \dim W.$$

Remark. The map π in the theorem is sometimes called the **(canonical) projection**.

Corollary 4. Let V, W be **finite dimensional** vector spaces and $T : V \rightarrow W$ be linear, then the following are equivalent:

1. T is one-to-one.
2. T is onto.

3. $\dim V = \dim T$.

Note that the corollary does NOT hold for V, W being infinite dimensional (even if their dimension share the same cardinality):

Example 5. Let $D : F[x] \rightarrow F[x]$ be the formal derivative operator on $F[x]$, that is

$$D(a_n x^n + \cdots + a_0) := n a_n x^{n-1} + \cdots + a_1.$$

D is a linear transformation and D is onto, but D is not one to one.

Proof. Check it yourself that D is linear. Given $f \in F[x]$, f can be written as

$$\sum_{i=0}^n a_i x^i \quad a_i \in F.$$

Then we define

$$g = \sum_{j=1}^{n+1} \frac{a_{j-1}}{j} x^j \in F[x]$$

Compute that

$$D(g) = D\left(\sum_{j=1}^{n+1} \frac{a_{j-1}}{j} x^j\right) = \sum_{j=1}^{n+1} j \cdot \frac{a_{j-1}}{j} x^{j-1} = f.$$

Therefore, D is onto. Since $D(a) = 0$ for all $a \in F$, $\ker D \supsetneq \{0\}$, that is, D is not one to one. $\square$

Example 6. Let $F^{\mathbb{N}}$ be the space of sequences in F , it is an infinite-dimensional vector space over F . Define the right shift operator $\phi : F^{\mathbb{N}} \rightarrow F^{\mathbb{N}}$ by

$$\phi(a_1, a_2, \cdots) = (0, a_1, a_2, \cdots)$$

Then ϕ is linear, ϕ is one-to-one, but ϕ is not onto.

Proof. Check yourself that ϕ is linear. We first prove that ϕ is one to one, let $(b_1, b_2, \cdots) \in \ker \phi$, then

$$\phi(b_1, b_2, \cdots) = (0, b_1, b_2, \cdots) = (0, 0, 0, \cdots),$$

which forces $b_1 = b_2 = \cdots = 0$. Therefore, $\ker \phi \subseteq \{0\}$, and naturally we have $\{0\} \subseteq \ker \phi$, thus ϕ is one to one.

ϕ is not onto since we cannot find a sequence $\{c_n\}$ such that

$$\phi(\{c_n\}) = (d_1, d_2, \cdots) \quad d_1 \neq 0.$$

Any sequence in $\text{im } D$ must have 0 in its first component. $\square$

Let W_1, W_2 be subspaces of V such that $V = W_1 \oplus W_2$, the projection on W_1 along W_2 is a linear transformation on V defined by

$$T(v) = w_1, \text{ where } v = w_1 + w_2 \text{ for unique } w_1 \in W_1, w_2 \in W_2.$$

Note that $T(w) = w$ for all $w \in W_1$ since w can be uniquely written as $w = w + 0$, therefore, for any $v \in V$, say $v = w_1 + w_2$ for $w_1 \in W_1, w_2 \in W_2$, then

$$T^2(v) := T(T(v)) = T(w_1) = w_1 = T(v)$$

That is, $T^2 = T$. In fact, every linear transformation T on V satisfying $T^2 = T$ must be a projection:

Theorem 7 (Characterization of projections). *Let V be a vector space and $T : V \rightarrow V$ be linear, then the following are equivalent:*

1. $T^2 = T$.
2. T is a projection on some subspace of V .

Proof. (1 $\implies$ 2) Suppose that $T^2 = T$, then decompose $v \in V$ into

$$v = T(v) + (v - T(v))$$

Since $T(v - T(v)) = T(v) - T(T(v)) = 0$, $v - T(v) \in \ker T$, we have $V = \operatorname{im} T + \ker T$ (why?). It suffices to prove that $\operatorname{im} T \cap \ker T = \{0\}$, suppose that $x \in \operatorname{im} T \cap \ker T$, then $x = T(y)$ for some $y \in V$ and $T(x) = 0$, therefore,

$$0 = T(x) = T(T(y)) = T(y) = x,$$

Therefore, we have $V = \ker T \oplus \operatorname{im} T$. Given $v \in V$, we can decompose $v = w_1 + w_2$ uniquely where $w_1 \in \ker T$ and $w_2 \in \operatorname{im} T$, then $T(v) = T(w_1) + T(w_2) = T(w_2)$, since $T^2(v) = T(v)$, we have $T(w_2) = w_2$. That is $T(v) = w_2$, so T is a projection on $\operatorname{im} T$ along $\ker T$.

(2 $\implies$ 1) We have proved this just before this theorem. □

Remark. In this case, V can be decomposed into $V = \operatorname{im} T \oplus \ker T$ and T is the projection on $\operatorname{im} T$ along $\ker T$. The converse does NOT hold: if we only assume $V = \operatorname{im} T \oplus \ker T$, then T is NOT necessarily a projection. For example, consider $V = \mathbb{R}^2$,

$$T : \mathbb{R}^2 \rightarrow \mathbb{R}^2 \quad T(x, y) = (2y, 0).$$

References

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